



3 – Discrete Random Variables

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Recap: PMF and CDF

Let X be a discrete random variable. The **probability mass function (PMF)** of X is defined as

$$p_X(x) = \mathbb{P}(X = x)$$

If X is a discrete random variable with PMF $p(x)$, then the **Cumulative Density Function (CDF)** of X is defined by

$$F(x) = \mathbb{P}(X \leq x) = \sum_{t \leq x} p(t), \quad -\infty < x < \infty$$

where the sum runs over all numbers $t \leq x$.

Recap: Probabilistic Complements

$$Pr(x \in A) = 1 - Pr(x \in \bar{A})$$

$$Pr(x \leq 20) = 1 - Pr(x > 20)$$

Recap: $E(X)$ & $\text{Var}(X)$

The **expected value** (or **mean**) of a **discrete** random variable X with PMF $p(x)$ is

$$\mathbb{E}[X] = \sum_x xp(x)$$

where the sum runs over all numbers x with $p(x) > 0$.

$$\mathbb{E}[g(X)] = \sum_x g(x)p(x).$$

Recap: $E(X)$ & $\text{Var}(X)$

Theorem 10 (Formula for Variance)

$$\text{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

The **standard deviation** of a random variable X is defined as

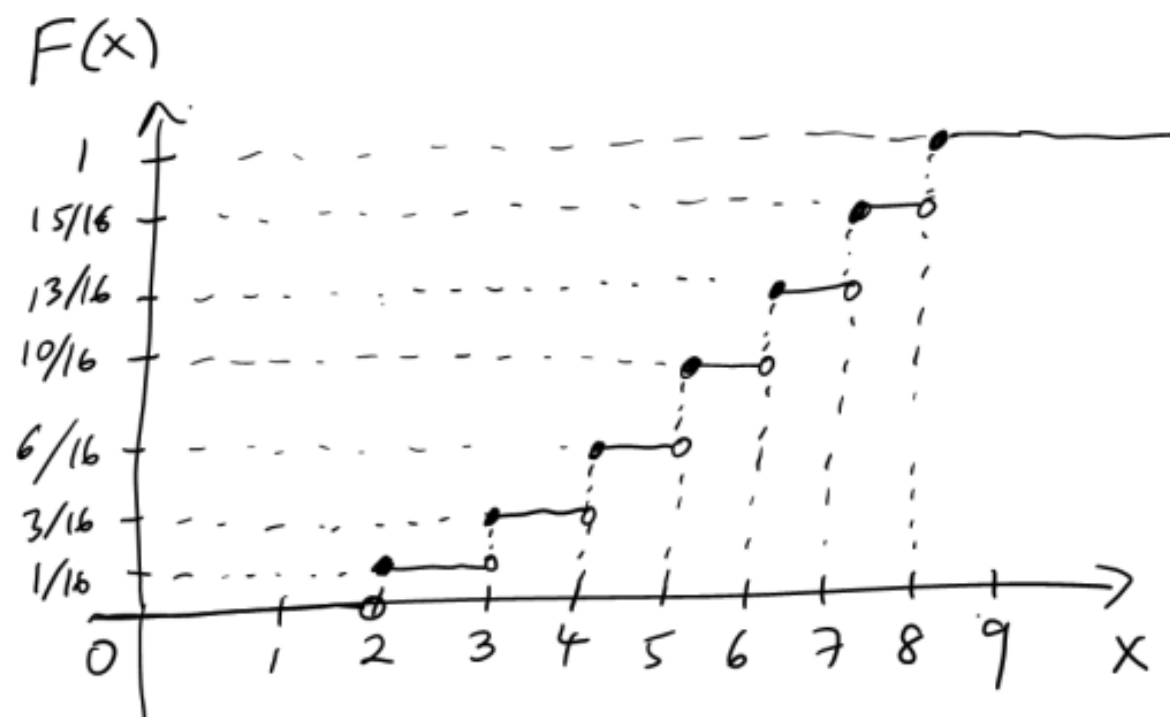
$$\sigma_X = \sqrt{\text{Var}(X)}.$$

Problem 1 (discrete random variables, PMF, CDF)

For the following random variables X and Y , compute their PMF, CDF, expected value, and variance. Draw graphs of the CDFs.

- (a) A fair 4-sided dice (with faces numbered 1 through 4) is rolled twice independently. Let X be the sum of the two numbers obtained.
- (b) Let a chip be taken at random from a bowl that contains six white chips, three red chips, and one blue chip. Let the random variable $X = 1$ if the outcome is a white chip, let $X = 5$ if the outcome is a red chip, and let $X = 10$ if the outcome is a blue chip.

$$F(x) = \begin{cases} 0, & x < 2 \\ \frac{1}{16}, & 2 \leq x < 3 \\ \frac{3}{16}, & 3 \leq x < 4 \\ \frac{6}{16}, & 4 \leq x < 5 \\ \frac{10}{16}, & 5 \leq x < 6 \\ \frac{13}{16}, & 6 \leq x < 7 \\ \frac{15}{16}, & 7 \leq x < 8 \\ 1, & x \geq 8 \end{cases}$$



– Expected value:

$$\begin{aligned}\mathbb{E}[X] &= \sum_x xp(x) \\ &= 2(1/16) + 3(2/16) + 4(3/16) + 5(4/16) + 6(3/16) + 7(2/16) + 8(1/16) \\ &= 5\end{aligned}$$

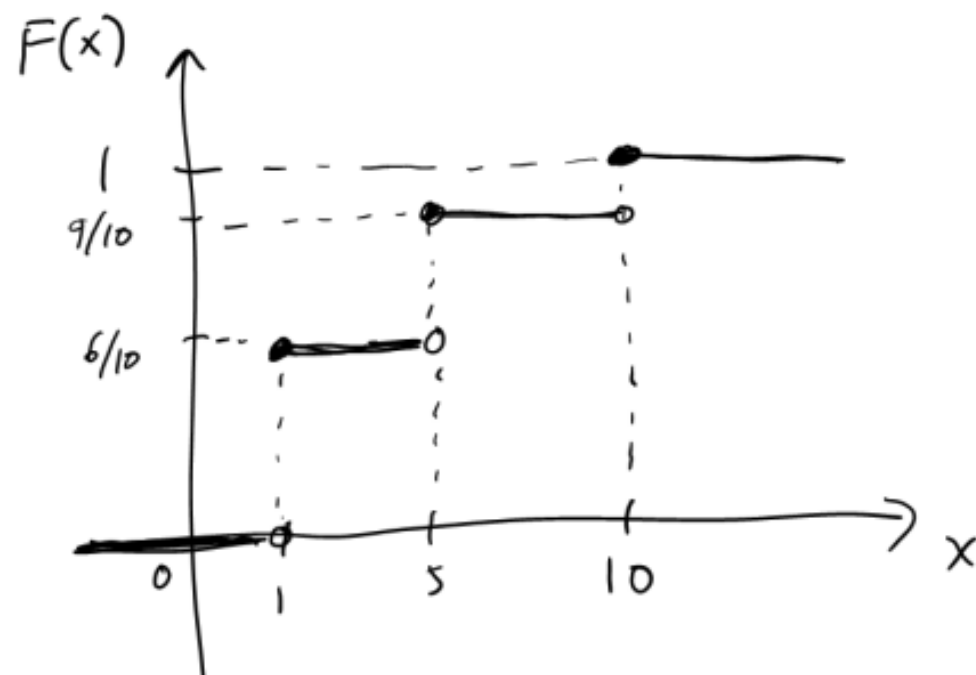
– Variance:

$$\begin{aligned}\mathbb{E}[X^2] &= \sum_x x^2p(x) \\ &= 2^2(1/16) + 3^2(2/16) + 4^2(3/16) + 5^2(4/16) + 6^2(3/16) + 7^2(2/16) + 8^2(1/16) \\ &= 27.5\end{aligned}$$

Hence,

$$\text{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = 27.5 - (5)^2 = 2.5.$$

$$F(x) = \begin{cases} 0, & x < 1 \\ \frac{6}{10}, & 1 \leq x < 5 \\ \frac{9}{10}, & 5 \leq x < 10 \\ 1, & x \geq 10 \end{cases}$$



– Expected value:

$$\begin{aligned}\mathbb{E}[X] &= \sum_x xp(x) \\ &= 1(6/10) + 5(3/10) + 10(1/10) \\ &= 3.1\end{aligned}$$

– Variance:

$$\begin{aligned}\mathbb{E}[X^2] &= \sum_x x^2p(x) \\ &= 1^2(6/10) + 5^2(3/10) + 10^2(1/10) \\ &= 18.1\end{aligned}$$

Hence,

$$\text{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = 18.1 - (3.1)^2 = 8.49.$$

Problem 2 (linearity of expected values)

Find the expected value of the sum obtained when

- (a) 10 fair dice are rolled.
- (b) n fair dice are rolled.

Problem 3 (linearity of expected values)

A fair four-sided die has two faces numbered 0 and two faces numbered 2. Another fair four-sided die has its faces numbered 0, 1, 4, and 5. The two dice are rolled. Let X and Y be the respective outcomes of the roll. Find the expected value of $X + Y$.

Problem 4 (expected values, variance)

Let X equal the larger outcome when a pair of four-sided dice (with faces numbered from 1 through 4) is rolled. If both dice give the same number, then X is equal to that number. Find the expected value, variance and standard deviation of X .

Recap: Geometric

Theorem 15 (Geometric distribution)

If $X \sim \text{Geom}(p)$, then

$$\text{PMF: } p(x) = (1 - p)^{x-1} p, \quad x = 1, 2, \dots,$$

$$\mathbb{E}[X] = \frac{1}{p}, \quad \text{Var}[X] = \frac{1 - p}{p^2}.$$

SolutionThe PMF of X is given as follows:

x	1	2	3	4
$f(x)$	$\frac{1}{16}$	$\frac{3}{16}$	$\frac{5}{16}$	$\frac{7}{16}$

$$\text{expected value} = \mathbb{E}[X] = 1 \cdot \frac{1}{16} + 2 \cdot \frac{3}{16} + 3 \cdot \frac{5}{16} + 4 \cdot \frac{7}{16} = 3.125.$$

$$\mathbb{E}[X^2] = 1^2 \cdot \frac{1}{16} + 2^2 \cdot \frac{3}{16} + 3^2 \cdot \frac{5}{16} + 4^2 \cdot \frac{7}{16} = 10.625.$$

$$\text{variance} = \text{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = 10.625 - 3.125^2 = 0.8594.$$

$$\text{standard deviation} = \sqrt{\text{Var}[X]} = \sqrt{0.8594} = 0.9270.$$

Problem 5 (Discrete random variables: Bernoulli, Binomial and Geometric)

Apples are packaged automatically in 3-pound bags. Suppose that 4% of the time the bag of apples weighs less than 3 pounds. If you select bags randomly and weigh them in order to discover one underweight bag of apples, find the probability that the number of bags that must be selected is

- (a) exactly 20 You stop selecting new bags the moment you discover one underweight bag of apples. ie. The X-th bag you selected is underweight.
- (b) at least 20

You may assume the following formula for a geometric sum with ratio r : $\sum_{i=1}^n r^{i-1} = \frac{1-r^n}{1-r}$.

Recap: Bernoulli

$$p(x) = \begin{cases} 1 - p & \text{if } x = 0 \\ p & \text{if } x = 1 \\ 0 & \text{otherwise.} \end{cases}$$

Theorem 11 (Bernoulli)

If $X \sim \text{Bernoulli}(p)$, then

$$\mathbb{E}[X] = p, \quad \text{Var}[X] = p(1 - p).$$

It follows that the standard deviation of X is $\sqrt{p(1 - p)}$.

Recap: Binomial

Theorem 12 (Binomial distribution)

If $X \sim \text{Binomial}(n, p)$, then

$$\text{PMF: } p(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n.$$

$$\mathbb{E}[X] = np, \quad \text{Var}[X] = np(1-p).$$

Problem 6 (Discrete random variables: Bernoulli, Binomial and Geometric)

It is believed that approximately 75% of American youth now have insurance due to the health care law. Suppose this is true, and let X equal the number of American youth in a random sample of $n = 15$ with health insurance.

- (a) How is X distributed?
- (b) Find the probability that X is at most 13.
- (c) Give the mean, variance, and standard deviation of X .

Problem 7 (Discrete random variables: Bernoulli, Binomial and Geometric)

Your stockbroker is free to take your calls about 60% of the time; otherwise, he is talking to another client or is out of the office. You call him at five random times during a given month. (Assume independence.)

- (a) What is the probability that he will accept exactly three of your five calls?
- (b) What is the probability that he will accept at least one of the calls?

Problem 8 (Discrete random variables: Bernoulli, Binomial and Geometric)

It is known that 2% of people whose luggage is screened at an airport have questionable objects in their luggage.

- (a) What is the probability that a string of 15 people pass through screening successfully before an individual is caught with a questionable object?
- (b) What is the expected number of people to pass through before an individual is caught with a questionable object?